

Deepak Mallampati

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Kent, OH -- F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) who builds production AI systems and deploys them to non-technical domain users. Track record of shipping end-to-end: RAG systems (~35% retrieval precision gain), agentic LLM workflows (>30% hallucination reduction), and predictive ML pipelines (15-20% recall improvement on high-risk categories) -- all in HIPAA-conscious healthcare environments where guardrails and user trust matter more than raw model accuracy. Independently shipped **Manga Lens** (live Chrome extension, 4 AI providers, 29 sites) and co-built **E-Farming** (80-120 active users). Comfortable owning the full loop: discovery with domain stakeholders through code, deployment, monitoring, and iteration.

CORE COMPETENCIES

Forward Deployment • Agentic LLM Workflows • RAG Systems • Python / FastAPI / Flask • TypeScript / React • AI Workflow Automation • LangChain / LlamaIndex • Docker / CI/CD • Non-technical User Tooling • End-to-end Product Ownership • scikit-learn / XGBoost • Production ML Deployment

EXPERIENCE

Progress Solutions Inc.

Jul 2025 -- Present

Jr. AI/ML Engineer Trainee — USA | Healthcare Technology, AI Consulting

- Built and deployed **RAG systems** for clinical knowledge retrieval; recursive semantic chunking + transformer embeddings improved retrieval precision by ~35% and reduced irrelevant retrieval by >30%; response alignment >90% in evaluation.
- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care navigation, documentation) with structured reasoning and grounding rules; improved agent stability by ~25%.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services with Docker; structured logging and load simulation reduced post-deployment defects by ~30%.
- Produced stakeholder-facing system-limitation documentation for non-technical clinical staff; enabled confident, audit-ready AI usage across teams with no ML background.
- Shipped predictive ML pipelines (scikit-learn, XGBoost) for patient no-show and care engagement scoring; improved recall by 15-20% on high-risk categories.

Suvidha Foundation

Jan 2022 -- Mar 2022

Machine Learning Engineer — Hyderabad, India | Nonprofit

- Built transformer-based video summarization for 5,000+ sessions; reduced manual review time by 60-70% for non-technical staff.
- Implemented document Q&A with RAG; reduced hallucinated outputs by ~30% and raised answer accuracy >85%.
- Deployed as a Flask API with a web interface designed for non-technical users -- same audience type as childcare providers.

Energy Solutions International (Emerson)

Jun 2022 -- Apr 2023

Database & DevOps Performance Engineer (Intern) — Hyderabad, India

- Built CI/CD pipelines with Jenkins; reduced deployment errors >30% and improved release cycle by ~35-40%.
- Designed Grafana observability dashboards for production SQL systems; cut incident recurrence ~25%.

PROJECTS

Manga Lens shipped -- Chrome Web Store

Solo-built and shipped a Chrome extension integrating **4 AI vision providers** (Claude, GPT-4o, Ollama) for real-time manga/webtoon translation. Manifest V3, content scripts, multi-section capture pipeline with viewport-slice screenshots. Seven-day translation cache, per-domain configs for 29 sites, live users. Full product ownership: design, code, privacy policy, Chrome review, user feedback loop.

Python, TypeScript, Chrome APIs, Claude/GPT-4o/Ollama, Manifest V3

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline (Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with schema-validated JSON contracts between agents to prevent cascading hallucinations. RAG-grounded CPT/ICD validation; explainable risk scoring with audit-ready reasoning traces.

Python, LangChain, vector search, FastAPI, Docker

E-Farming Digital Marketplace founder

Independently designed and shipped a PHP/MySQL/Bootstrap AgriTech marketplace connecting small-scale farmers with buyers. Onboarded **80-120 active users** in phase one through direct farmer outreach; iterated on cart, reviews, and community blogging based on user feedback.

PHP, MySQL, Bootstrap -- full-cycle owner: discovery, build, launch, user feedback

Dream Decoder -- AI Multimodal Platform

Full-stack FastAPI + React/TypeScript app with coordinated multimodal APIs (Perplexity Sonar, Replicate). Intermediate prompt transformation layers improved contextual alignment **~30%** and first-pass image success by **~25-30%**.

FastAPI, React, TypeScript, Perplexity Sonar, Replicate

EDUCATION

Master's Degree — Kent State University, Ohio, USA

Aug 2023 -- May 2025

TECHNICAL SKILLS

Languages & Frameworks: Python, TypeScript, JavaScript, React, FastAPI, Flask, SQL (PostgreSQL, T-SQL)

AI/LLM & Automation: LLM APIs (Claude, GPT-4o, Gemini, Ollama), RAG, agentic workflows, LangChain, LlamaIndex, prompt engineering, structured outputs, evaluation pipelines

Developer Tooling: Cursor, Claude Code, Docker, CI/CD (Jenkins), Grafana, RESTful APIs, MLOps/LLMOps

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, YOLOv8

Work Authorization: F-1 OPT -- authorized to work in the United States now; future sponsorship required